THE VARIATION THRESHOLD, KEMENY'S CONSTANT AND STATIONARY TIMES OF REVERSIBLE MARKOV CHAINS

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Abstract. For finite reversible Markov chains, Aldous and Fill asked whether the variation threshold τ_1 is at most the average hitting time τ_0 (Kemeny's constant), and whether the relaxation time τ_2 is at most $\tau_1^{(2)}$, the largest over initial states of the least mean of a stopping time at which the chain has its stationary distribution. The answer to the first question is no: a birth and death chain on three states with a strong drift has $\tau_1 > \tau_0$, and on the analogous chains with $k+1$ states the ratio τ_1/τ_0 tends to t_k/k as the drift grows, where t_k is the point at which the survival function of the Gamma$(k,1)$ distribution equals e^{-1}; for $k=4$ this limit is $1.0879\ldots$. The answer to the second question is yes, in continuous and in discrete time.

1. Introduction

Let X be an irreducible Markov chain in continuous time on a finite set I of $n \geq 2$ states, with generator $Q = (q_{ij})$ and stationary distribution π, and assume that X is reversible: $\pi_i q_{ij} = \pi_j q_{ji}$ for all i, j. Write $\mathbb{P}_i$ and $\mathbb{E}_i$ for the chain started at i, $T_j = \inf\{t \geq 0 : X_t = j\}$ for the hitting time of j, $\|\mu - \nu\| = \max_A |\mu(A) - \nu(A)|$ for the variation distance between probability measures on I, $\mu(h) = \sum_k \mu_k h(k)$, and $0 = \lambda_1 < \lambda_2 \leq \cdots \leq \lambda_n$ for the eigenvalues of $-Q$. In the notation of Aldous and Fill [1, Chapter 4], the average hitting time, the variation threshold and the relaxation time are

$$\tau_0 = \sum_{i,j} \pi_i \pi_j \mathbb{E}_i T_j, \qquad \tau_1 = \min\{t \geq 0 : \bar{d}(t) \leq e^{-1}\}, \qquad \tau_2 = \frac{1}{\lambda_2},$$

where $\bar{d}(t) = \max_{i,j} \|\mathbb{P}_i(X_t \in \cdot) - \mathbb{P}_j(X_t \in \cdot)\|$. By the random target lemma [1, Corollary 2.14], $\sum_j \pi_j \mathbb{E}_i T_j$ does not depend on i, so τ_0 is this common value, often called Kemeny's constant [4]. We also need

$$\tau_1^{(2)} = \max_i \inf_U \mathbb{E}_i U,$$

where the infimum is over randomized stopping times U with $\mathbb{P}_i(X_U \in \cdot) = \pi$. A randomized stopping time is a stopping time for a filtration $(\mathcal{F}_t)$ with respect to which X is a Markov process, for instance the filtration generated by X and a random variable independent of X. Stopping times that achieve a given distribution in least mean are studied by Lovász and Winkler [5].

The comparison of these parameters goes back to Aldous [2]. Among the inequalities between them proved in [1, Chapter 4] are

$$\tau_2 \leq \tau_1 \leq 66\,\tau_0, \qquad \tau_2 \leq \tau_0, \qquad \tau_1^{(2)} \leq \tau_0$$

(Lemmas 4.23, 4.15 and 4.24 and Section 4.3.2 there), and by [1, Theorem 4.6] the parameter $\tau_1^{(2)}$ belongs to a family of parameters equivalent to τ_1, that is, bounded above and below by absolute constant multiples of it. Aldous and Fill chose τ_1 as the representative of this family

Date: 13 October 2026; revised 12 November 2026.

2020 *Mathematics Subject Classification.* 60J27, 60J10.

Key words and phrases. Reversible Markov chain, mixing time, Kemeny's constant, relaxation time, stationary time.

partly because it gives the constant 1 in $\tau_2 \le \tau_1$, and they ask whether the constants in other basic inequalities relating the family to other parameters can be made 1 as well. Their Open Problem 4.48 asks:

(a) Is $\tau_1 \le \tau_0$?
(b) Is $\tau_2 \le \tau_1^{(2)}$?

We show that the answer to (a) is no and the answer to (b) is yes. Up to an absolute constant factor, τ_2 is the largest expected hitting time of a set A with $\pi(A) \ge 1/2$ for the chain started from π conditioned on the complement of A; this is a theorem of Hermon [3, Theorem 1.1], which resolves [1, Open Problem 4.38].

For $k \ge 1$ and $\varepsilon > 0$ let $X^{(k,\varepsilon)}$ be the chain on $\{0, 1, \dots, k\}$ that jumps from i to $i-1$ at rate 1 for $1 \le i \le k$ and from i to $i+1$ at rate ε for $0 \le i \le k-1$. It is reversible, with π_i proportional to ε^i. Let Γ_k be a random variable with the Gamma$(k, 1)$ distribution, the law of a sum of k independent standard exponential variables, and let t_k be the solution of

$$\mathbb{P}(\Gamma_k > t_k) = e^{-t_k} \sum_{j=0}^{k-1} \frac{t_k^j}{j!} = e^{-1}. \tag{1.1}$$

Theorem 1. (i) *If $0 < \varepsilon \le 1/100$, then $X^{(2,\varepsilon)}$ has*

$$\tau_0 = \frac{2(1+\varepsilon)}{1+\varepsilon+\varepsilon^2} < 2 < \tau_1.$$

(ii) *For every $k \ge 1$, the chain $X^{(k,\varepsilon)}$ has $\tau_0 \to k$ and $\tau_1 \to t_k$ as $\varepsilon \to 0$.*

As $\varepsilon \to 0$ the chain $X^{(k,\varepsilon)}$ becomes the pure death chain absorbed at 0. For that chain $\bar{d}(t) = \mathbb{P}(\Gamma_k > t)$, the probability that the chain started at k has not reached 0 by time t, while τ_0 tends to k, the mean of Γ_k; so $\tau_1 > \tau_0$ for small ε whenever $\mathbb{P}(\Gamma_k > k) > e^{-1}$, which holds for $k = 2$ because $\mathbb{P}(\Gamma_2 > 2) = 3e^{-2} = 0.406\ldots$. For $\varepsilon = 1/100$ the chain $X^{(2,\varepsilon)}$ has $\tau_1 = 2.1330\ldots$ and $\tau_1/\tau_0 = 1.0666\ldots$ (Section 4). The limits t_k/k in part (ii) are 1, $1.0730\ldots$, $1.0860\ldots$, $1.0879\ldots$ and $1.0868\ldots$ for $k = 1, \dots, 5$; they tend to 1 as $k \to \infty$ by the law of large numbers, and among $k \le 60$ the largest is $t_4/4 = 1.08799\ldots$. So the best constant C such that $\tau_1 \le C\tau_0$ for every finite reversible chain satisfies $t_4/4 \le C \le 66$. We do not know its value.

Theorem 2. *Let u be a real function on I with $Qu = -\lambda_2 u$ and $\max_k |u(k)| = 1$. For every state i and every randomized stopping time U with $\mathbb{P}_i(X_U \in \cdot) = \pi$,*

$$\mathbb{E}_i U \ge \tau_2 \, |u(i)|.$$

In particular $\tau_2 \le \tau_1^{(2)}$.

The constant 1 cannot be lowered: two-state chains have $\tau_2 = \tau_1^{(2)}$ (Remark 1 in Section 3). Since $\tau_1^{(2)} \le \tau_0$, Theorem 2 refines the inequality $\tau_2 \le \tau_0$.

Both theorems hold for chains in discrete time with the conventions of [1, (3.39) and Section 4.3.1]. For an irreducible reversible transition matrix P with second largest eigenvalue λ_2, the relaxation time is $\tau_2 = 1/(1-\lambda_2)$; τ_0 and $\tau_1^{(2)}$ are defined as above, with hitting times and stopping times taking values in $\{0, 1, 2, \dots\}$; and τ_1 is the variation threshold of the continuized chain, the chain in continuous time with generator $P - I$. Section 2.3 shows that the chains $I + cQ$, with Q the generator of $X^{(k,\varepsilon)}$ and $0 < c \le 1/(1+\varepsilon)$, have the same ratio τ_1/τ_0 as $X^{(k,\varepsilon)}$, and that for $k = 2$, $0 < \varepsilon \le 1/100$ and small c the variation threshold of the discrete chain itself also exceeds τ_0; Section 3 proves Theorem 2 in discrete time, with $Pu = \lambda_2 u$ in place of $Qu = -\lambda_2 u$.

2. Proof of Theorem 1

Write Q_ε for the generator of $X^{(k,\varepsilon)}$ and $\bar{d}_\varepsilon$ for its function $\bar{d}$. Detailed balance, $\pi_i\varepsilon = \pi_{i+1}$, shows that $X^{(k,\varepsilon)}$ is reversible with $\pi_i \propto \varepsilon^i$. By the random target lemma and the eigentime identity [1, Proposition 3.13],

$$\tau_0 = \sum_{m\ge 2} \frac{1}{\lambda_m}. \tag{2.1}$$

2.1. **The chain on three states.** Let $k = 2$. The matrix $-Q_\varepsilon$ has trace $2 + 2\varepsilon$, principal 2×2 minors ε^2, ε and 1, and determinant 0. So $\lambda_2 + \lambda_3 = 2 + 2\varepsilon$ and $\lambda_2\lambda_3 = 1 + \varepsilon + \varepsilon^2$, and by (2.1)

$$\tau_0 = \frac{\lambda_2 + \lambda_3}{\lambda_2\lambda_3} = \frac{2(1+\varepsilon)}{1+\varepsilon+\varepsilon^2} < 2.$$

Now fix $t \in [0, 2]$. Comparing the initial states 0 and 2 on the event $\{X_t = 0\}$ gives

$$\bar{d}_\varepsilon(t) \ge \mathbb{P}_0(X_t = 0) - \mathbb{P}_2(X_t = 0). \tag{2.2}$$

The chain started at 0 stays there for an exponential time with rate ε, so

$$\mathbb{P}_0(X_t = 0) \ge e^{-\varepsilon t}. \tag{2.3}$$

For the chain started at 2, let $Y_0 = 2, Y_1 = 1, Y_2, \dots$ be its jump chain and $S_0, S_1, \dots$ its holding times. Given the jump chain, the holding times are independent, and S_m is exponential with rate 1 if $Y_m = 2$, with rate $1 + \varepsilon$ if $Y_m = 1$ and with rate ε if $Y_m = 0$. Let $N = \min\{m : Y_m = 0\}$. Then $N \ge 2$, $Y_{N-1} = 1$ and

$$T_0 = S_0 + \cdots + S_{N-1} \ge S_0 + S_{N-1}.$$

Given the jump chain, S_0 and S_{N-1} are independent and exponential with rates 1 and $1 + \varepsilon$; since these conditional laws do not depend on the jump chain, S_0 and S_{N-1} are independent with these laws. Let E and E' be independent standard exponential variables and

$$g(s) = \mathbb{P}(E + E' > s) = e^{-s}(1 + s), \qquad s \ge 0.$$

Since $E + E'/(1+\varepsilon) \ge (E + E')/(1+\varepsilon)$,

$$\mathbb{P}_2(X_t = 0) \le \mathbb{P}_2(T_0 \le t) \le \mathbb{P}\Big(E + \frac{E'}{1+\varepsilon} \le t\Big) \le \mathbb{P}(E + E' \le (1+\varepsilon)t) = 1 - g\big((1+\varepsilon)t\big). \tag{2.4}$$

The function g is nonincreasing and $(1+\varepsilon)t \le 2 + 2\varepsilon$, so (2.2), (2.3), (2.4), $e^{-\varepsilon t} \ge 1 - 2\varepsilon$ and $e^{-2\varepsilon} \ge 1 - 2\varepsilon$ give

$$\bar{d}_\varepsilon(t) \ge e^{-\varepsilon t} - 1 + g\big((1+\varepsilon)t\big) \ge g(2 + 2\varepsilon) - 2\varepsilon \ge e^{-2}(1 - 2\varepsilon)(3 + 2\varepsilon) - 2\varepsilon. \tag{2.5}$$

The right side of (2.5) exceeds e^{-1} if and only if

$$f(\varepsilon) = 3 - e - (4 + 2e^2)\varepsilon - 4\varepsilon^2 > 0.$$

The function f is decreasing on $[0, \infty)$, and $e < 2.7183$ and $e^2 < 7.3891$ give $f(1/100) > 0.2817 - 0.1878 - 0.0004 > 0.09$. Hence, when $0 < \varepsilon \le 1/100$, $\bar{d}_\varepsilon(t) > e^{-1}$ for every $t \in [0, 2]$, and $\tau_1 > 2$. This proves part (i).

2.2. **Limits.** Let $k \geq 1$ and $\varepsilon \to 0$. The matrix Q_ε converges to the matrix Q_0 with the same entries at $\varepsilon = 0$, the generator of the pure death chain that moves from i to $i-1$ at rate 1 and is absorbed at 0. The matrix $-Q_0$ is lower triangular with diagonal $(0, 1, \ldots, 1)$, so its eigenvalues are 0, which is simple, and 1, with multiplicity k. The eigenvalues of a matrix depend continuously on its entries, and 0 is an eigenvalue of every $-Q_\varepsilon$; so for small ε the eigenvalue 0 is the only eigenvalue of $-Q_\varepsilon$ near 0, and the others, $\lambda_2, \ldots, \lambda_{k+1}$, tend to 1. By (2.1), $\tau_0 \to k$.

For τ_1, let $P_t^\varepsilon = e^{tQ_\varepsilon}$ for $\varepsilon \geq 0$. The identity $e^{tA} - e^{tB} = \int_0^t e^{(t-s)A}(A-B)e^{sB}\,ds$ shows that $P_t^\varepsilon \to P_t^0$ uniformly for t in compact sets, and hence $\bar{d}_\varepsilon \to \bar{d}_0$ uniformly on compact sets, where $\bar{d}_0(t)$ is defined from P_t^0 in the same way. Under Q_0 the chain started at i passes through $i-1, \ldots, 1$ to 0 with independent standard exponential holding times and then stays at 0, so $P_t^0(0, \cdot)$ is the point mass at 0 and $P_t^0(j, 0) = \mathbb{P}(\Gamma_j \leq t)$, where $\Gamma_0 = 0$. Let i, j be states, $\mu = P_t^0(i, \cdot)$ and $\nu = P_t^0(j, \cdot)$, and suppose that $\mu(0) \geq \nu(0)$. Then

$$\|\mu - \nu\| = \sum_x (\mu(x) - \nu(x))^+ \leq \mu(0) - \nu(0) + \sum_{x \neq 0} \mu(x) = 1 - \nu(0) = \mathbb{P}(\Gamma_j > t) \leq \mathbb{P}(\Gamma_k > t),$$

with equality throughout when $i = 0$ and $j = k$. Hence $\bar{d}_0(t) = \mathbb{P}(\Gamma_k > t)$. This function is continuous and strictly decreasing from 1 to 0. For $0 < \delta < t_k$ it is therefore at least $\bar{d}_0(t_k - \delta) > e^{-1}$ on $[0, t_k - \delta]$, and $\bar{d}_0(t_k + \delta) < e^{-1}$. By uniform convergence on $[0, t_k + \delta]$, for all small ε we have $\bar{d}_\varepsilon > e^{-1}$ on $[0, t_k - \delta]$ and $\bar{d}_\varepsilon(t_k + \delta) < e^{-1}$, so $t_k - \delta < \tau_1 \leq t_k + \delta$. This proves part (ii).

2.3. **Discrete time.** Let $0 < c \leq 1/(1+\varepsilon)$. Every diagonal entry of Q_ε is at least $-(1+\varepsilon)$, so $P = I + cQ_\varepsilon$ is a transition matrix, irreducible and reversible with respect to the same π. Its continuization has generator $P - I = cQ_\varepsilon$ and is $X^{(k,\varepsilon)}$ run at speed c. Write $\tau_0(P)$ and $\tau_1(P)$ for the parameters of P, and τ_0 and τ_1 for those of $X^{(k,\varepsilon)}$. The variation threshold of the continuization, which is $\tau_1(P)$, equals τ_1/c. Continuization does not change mean hitting times [1, Section 4.3.2], so $\tau_0(P) = \tau_0/c$. Hence $\tau_1(P)/\tau_0(P) = \tau_1/\tau_0$: for $k = 2$ and $0 < \varepsilon \leq 1/100$ the chain P has $\tau_1(P) > \tau_0(P)$, and its ratio tends to t_k/k as $\varepsilon \to 0$.

For $k = 2$ the inequality also holds, once c is small, for the variation threshold of P itself, $\tau_1^{\mathrm{disc}} = \min\{m \geq 0 : \bar{d}(m) \leq e^{-1}\}$, where $\bar{d}(m)$ is computed from the matrix P^m [1, Section 4.3.1]. Let $0 < \varepsilon \leq 1/100$ and $Q = Q_\varepsilon$, and write $\|A\| = \max_i \sum_j |A_{ij}|$. The matrices $I + cQ$ and e^{cQ} are stochastic, so

$$\|(I + cQ)^m - e^{mcQ}\| \leq m\,\|I + cQ - e^{cQ}\| \leq m\,\frac{(c\|Q\|)^2}{2}\,e^{c\|Q\|},$$

which is at most $c\|Q\|^2 e^{c\|Q\|}$ when $mc \leq 2$. The function $\bar{d}$ of a stochastic matrix changes by at most $\|A - B\|$ when the matrix B is replaced by A, and by (2.5) and the bound $f(\varepsilon) \geq f(1/100) > 0.09$, $\bar{d}_\varepsilon(t) - e^{-1} \geq e^{-2} f(\varepsilon) > 0.01$ for $t \in [0, 2]$. Hence, when $c\|Q\|^2 e^{c\|Q\|} \leq 0.01$, the matrix P^m has $\bar{d} > e^{-1}$ for every $m \leq 2/c$, and $\tau_1^{\mathrm{disc}} > 2/c > \tau_0/c$.

3. Proof of Theorem 2

Proof of Theorem 2. Replacing u by $-u$ we may assume $u(i) \geq 0$. Since $\pi Q = 0$, we have $-\lambda_2 \pi(u) = \pi(Qu) = 0$, so $\pi(u) = 0$. Let U be a stopping time for a filtration $(\mathcal{F}_t)$ with respect to which X is Markov, with $\mathbb{P}_i(X_U \in \cdot) = \pi$. If $\mathbb{E}_i U = \infty$ there is nothing to prove, so assume that $\mathbb{E}_i U < \infty$. For every function h on I, the process $h(X_t) - h(X_0) - \int_0^t (Qh)(X_s)\,ds$ is an $(\mathcal{F}_t)$-martingale, so optional stopping at the bounded time $U \wedge t$, with $h = u$, gives

$$\mathbb{E}_i u(X_{U \wedge t}) - u(i) = -\lambda_2\,\mathbb{E}_i \int_0^{U \wedge t} u(X_s)\,ds.$$

Let $t \to \infty$. Since $U < \infty$ almost surely and $|u| \le 1$, the left side tends to $\mathbb{E}_i u(X_U) - u(i) = \pi(u) - u(i) = -u(i)$ by bounded convergence, and the right side tends to $-\lambda_2 \mathbb{E}_i \int_0^U u(X_s)\, ds$ by dominated convergence, since the integral is at most U in absolute value. Therefore

$$u(i) = \lambda_2\, \mathbb{E}_i \int_0^U u(X_s)\, ds \le \lambda_2\, \mathbb{E}_i U,$$

that is, $\mathbb{E}_i U \ge u(i)/\lambda_2 = \tau_2 |u(i)|$. Taking i with $|u(i)| = 1$ gives $\inf_U \mathbb{E}_i U \ge \tau_2$, and so $\tau_1^{(2)} \ge \tau_2$. □

Discrete time. Let P be an irreducible reversible transition matrix, $\lambda_2 < 1$ its second largest eigenvalue, and u a real function with $Pu = \lambda_2 u$, $\max_k |u(k)| = 1$ and $u(i) \ge 0$. Then $\pi(u) = \pi(Pu) = \lambda_2 \pi(u)$, so $\pi(u) = 0$, and $u(X_n) - u(X_0) + (1-\lambda_2) \sum_{m<n} u(X_m)$ is a martingale. The argument above, with sums in place of integrals, gives

$$u(i) = (1-\lambda_2)\, \mathbb{E}_i \sum_{m<U} u(X_m) \le (1-\lambda_2)\, \mathbb{E}_i U,$$

so $\mathbb{E}_i U \ge \tau_2 |u(i)|$ with $\tau_2 = 1/(1-\lambda_2)$, and $\tau_2 \le \tau_1^{(2)}$.

Remarks. 1. For the chain on $\{1,2\}$ that jumps from 1 to 2 at rate a and from 2 to 1 at rate b, $-Q$ has eigenvalues 0 and $a+b$, so $\tau_2 = 1/(a+b)$, and $\pi = (b,a)/(a+b)$. Started at 1, the stopping time that equals 0 with probability $b/(a+b)$ and T_2 otherwise, the choice being independent of the chain, has X_U distributed as π and mean $\frac{a}{a+b} \cdot \frac{1}{a} = \frac{1}{a+b}$, and similarly from 2. With Theorem 2 this gives $\tau_1^{(2)} = 1/(a+b) = \tau_2$. For this chain also $\bar{d}(t) = e^{-(a+b)t}$ and $\tau_1 = \tau_0 = 1/(a+b)$.

2. Started at i, run the chain until it hits a random state with law π chosen independently of the chain. This stopping time has X_U distributed as π and mean $\sum_j \pi_j \mathbb{E}_i T_j = \tau_0$, so $\tau_1^{(2)} \le \tau_0$ [1, Section 4.3.2]. With Theorem 2, $\tau_2 \le \tau_1^{(2)} \le \tau_0$, a refinement of the inequality $\tau_2 \le \tau_0$ of [1, Lemma 4.24].

3. Let $Z_{ij} = \int_0^\infty (\mathbb{P}_i(X_t = j) - \pi_j)\, dt$. By [1, (4.47)], $\tau_1^{(2)} = \max_{i,j}(-Z_{ij}/\pi_j)$, so Theorem 2 says that $\max_{i,j}(-Z_{ij}/\pi_j) \ge 1/\lambda_2$. This can also be seen directly. Take u as in Theorem 2, replaced by $-u$ if necessary so that $u(i) = 1$ for some state i. Since $\pi(u) = 0$, $\sum_j Z_{ij} u(j) = \int_0^\infty e^{-\lambda_2 t} u(i)\, dt = \tau_2$. Let $m = \max_j(-Z_{ij}/\pi_j)$, so that $Z_{ij} + m\pi_j \ge 0$ for every j. Since $\sum_j Z_{ij} = 0$ and $\pi(u) = 0$,

$$\tau_2 = \sum_j (Z_{ij} + m\pi_j) u(j) \le \sum_j (Z_{ij} + m\pi_j) = m.$$

4. The proof of Theorem 2 uses reversibility only to know that λ_2 is real with a real eigenfunction. For any finite irreducible chain in continuous time, any eigenvalue $\lambda \ne 0$ of $-Q$ with eigenvector $u \in \mathbb{C}^I$, and any probability measure μ on I, the same argument applied to the real and imaginary parts of u shows that every randomized stopping time U with $\mathbb{P}_i(X_U \in \cdot) = \mu$ satisfies $|\lambda| \max_k |u(k)|\, \mathbb{E}_i U \ge |u(i) - \mu(u)|$. For $\mu = \pi$ this gives $\tau_1^{(2)} \ge 1/|\lambda|$.

5. In discrete time Theorem 2 does not extend to the absolute relaxation time $1/(1 - \max(\lambda_2, |\lambda_n|))$, where λ_n is the smallest eigenvalue of P. For

$$P = \begin{pmatrix} \delta & 1-\delta \\ 1-\delta & \delta \end{pmatrix}, \qquad 0 < \delta < \tfrac{1}{2},$$

$\lambda_2 = \lambda_n = 2\delta - 1$ and $\tau_2 = 1/(2(1-\delta))$. Stopping at time 0 with probability 1/2 and otherwise at the first visit to the other state gives a stopping time with X_U distributed as π and mean $1/(2(1-\delta))$, so $\tau_1^{(2)} = 1/(2(1-\delta))$ by Theorem 2, while the absolute relaxation time is $1/(2\delta)$.

4. Computations

The repository listed under Data availability contains a Python script, `checks.py`, with its output, and a Lean file, `Margin.lean`. For $X^{(2,\varepsilon)}$ with $\varepsilon = 1/100$ and $\varepsilon = 1/1000$, the script computes τ_0 from the closed form, from the eigentime identity and from the mean hitting times; computes $\bar{d}_\varepsilon(t)$ from the matrix exponential and from the spectral decomposition, in 40-digit arithmetic; checks $\bar{d}_\varepsilon(t) > e^{-1}$, (2.2), (2.3), the first inequality of (2.5) and the bound $e^{-\varepsilon t} - 1 + g((1+\varepsilon)t) > e^{-1}$ on a grid of step 0.001 on $[0,2]$, and (2.4) at the multiples of 0.05 in $[0,2]$; and finds τ_1 by bisection. At $\varepsilon = 1/100$, $\tau_0 = 1.99980\ldots$, $\tau_1 = 2.13306\ldots$ and the smallest value of $\bar{d}_\varepsilon$ on $[0,2]$ is $0.40240\ldots$; at $\varepsilon = 1/1000$, $\tau_1/\tau_0 = 1.07244\ldots$. At $\varepsilon = 1/20$, where $f(\varepsilon) < 0$ and the proof does not apply, the ratio τ_1/τ_0 is still $1.0421\ldots$.

For the chains on $\{0,\ldots,k\}$ the script computes t_k for $k \le 60$ by bisection, with a sign check on an interval of width $2\cdot 10^{-40}$ for $k \le 8$; checks $\bar{d}_0(t) = \mathbb{P}(\Gamma_k > t)$, attained at the pair $(0,k)$, for $k = 2,3,4,6$; and computes τ_1/τ_0 at $\varepsilon = 10^{-3}, 10^{-5}, 10^{-7}$ for $k = 2,3,4,6$, where it approaches t_k/k.

For Theorem 2 it draws 3000 random reversible chains with 2 to 8 states in continuous time and 3000 in discrete time. For each it computes $\tau_1^{(2)}$ from [1, (4.47)] and checks the inequality of Theorem 2 at every state. For 600 of the chains in each sample it also solves, at every state i, the linear program that minimizes the total mass of a measure $x \ge 0$ on I with $xG = \pi - \delta_i$, where $G = Q$ or $G = P - I$ (the expected occupation measure of a stopping time U with $\mathbb{P}_i(X_U \in \cdot) = \pi$ satisfies these constraints and has total mass $\mathbb{E}_i U$), checks that its value agrees with [1, (4.47)], and checks the mean hitting times against the fundamental matrix. The smallest value of $\tau_1^{(2)}/\tau_2$ in each sample is 1.000000 to six decimals. The script also checks Remark 1 and the example of Remark 5; in the discrete-time sample, the absolute relaxation time exceeds $\tau_1^{(2)}$ for 1584 of the 3000 chains.

The Lean file, checked with Lean 4 [6] (version 4.33.1) and Mathlib [7], proves that g is nonincreasing, that $f(\varepsilon) > 0$ for $0 < \varepsilon \le 1/100$, and that $e^{-\varepsilon t} - 1 + g((1+\varepsilon)t) > e^{-1}$ for $0 < \varepsilon \le 1/100$ and $0 \le t \le 2$, which is the real-variable part of the proof of Theorem 1(i).

These computations test the identities and inequalities used in the proofs; they are not part of them.

Acknowledgements and funding

The research presented in this paper was supported by the European Research Council (ERC) under the European Union's Horizon Europe research and innovation programme (grant agreement No. 101041711), by the Simons Foundation as part of the Collaboration on the Mathematical and Scientific Foundations of Deep Learning, by Heights Labs, by Convex Nexus Capital, by the Israel Science Foundation (grant number 2258/19), and by the Israel Science Foundation (ISF Grant 4101/25). Computational resources were provided in part by Google Cloud through its education credits program (credit 530550103).

This work was done by multiple AI agents, Claude (Anthropic) and Codex (OpenAI), connected to Math Brain [8] and working together inside the Brain framework [10], a shared, persistent memory in which the agents record claims, proofs, failed attempts and methods, retrieve one another's results, coordinate who works on what, and audit each other's work. For more on the details behind the Brain and Math Brain, see the Math Brain paper [9], the Math Brain repository [8] and the Brain framework repository [10].

Declaration of competing interest

The author holds equity in Heights Labs and in Convex Nexus Capital and has a management role at Convex Nexus Capital. These entities are included in the funding declaration above.

Declaration of generative AI and AI-assisted technologies

The paper comes out of Math Brain, an ongoing project on the use of AI systems for research mathematics. Open Problem 4.48 of [1] was among the open problems in probability on which the project's agents worked in September 2026. On 6 September 2026 an agent running in OpenAI's Codex wrote the optional stopping argument that proves Theorem 2. On 14 September 2026 Claude Fable 5.1 (Anthropic) found the chains of Theorem 1, sketched the proof of its part (ii), and wrote out the proof of Theorem 2. On 13 October 2026 Claude Opus 5.5 (Anthropic), working in Claude Code, proved part (i) of Theorem 1 for $0 < \varepsilon \leq 1/100$, completed the proof of part (ii), searched the literature, and wrote the remarks, the computer checks, the Lean file and the manuscript. The author directed the work and is responsible for the content of the paper and for decisions concerning its dissemination.

Data availability

No external data were used. The code and the Lean file are available at `https://github.com/yspennstate/variation-threshold-stationary-times`.

The paper is part of the Math Brain project, whose public repository is `https://github.com/yspennstate/math-brain-public` [8].

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